

IC 210: Probability, Statistics and Random Processes

Course Code	IC 210
Course Name	Probability, Statistics and Random Processes
L-T-P-C	2.5-0.5-0-3
Prerequisites	Consent of the faculty member
Intended for	B.Tech 1st year
Distribution	Core
Approval	2nd Adhoc Senate (May 2012), in the B.Tech Institute Core (common core courses) curriculum
Discontinued	17th Senate (Aug 2018)

Course Contents

- **Introduction to Probability (Theory of Gambling):** Definitions, scope and examples; Sample spaces and events; Axiomatic definition of Probability; Joint and conditional probabilities; Independence, total probability; Bayes' rule and applications [5 Lectures]
- **Random Variables (Dealing with Uncertainty):** Definition of random variables, continuous and discrete random variables; Cumulative distribution function (cdf) for discrete and continuous random variables; Probability mass function (pmf); Probability density function (pdfs) and properties; Jointly distributed random variables; Conditional and joint density distribution functions; Functions of a random variable; Expectation: mean, variance and moments of a random variablesw. [10 Lectures]
- **Distribution Functions (Fitting of a Function):** Some special distributions: Uniform, Exponential, Chi-square, Gaussian, Binomial, and Poisson Distributions; Moment-generating and characteristic functions and their applications; Chebysev inequality; Central limit theorem and its significance; Parameter estimation and confidence intervals for parameters; Regression; Hypothesis Testing. [11 Lectures]
- **Random Processes (Modeling of Chance):** Autocorrelation and autocovariance functions; Stationarity; Ergodicity; Correlation and covariance; White noise process and white noise sequence; Random walk, Markov Processes, Markov chains, Introduction to Queuing theory [10 Lectures]

Text Books

1. Sheldon M. Ross, **Introduction to Probability and Statistics for Engineers and Scientists**, Academic Press, 2009
2. Kishor S. Trivedi, **Probability and Statistics with Reliability Queuing and Computer Science Applications**, 2nd Edition, Wiley-Interscience, 2001.

References

1. Athanasios Papoulis, **Probability Random Variables and Stochastic Process**, 4th edition, McGraw Hill, 2002.
2. D. C. Montgomery and G. C. Runger, **Applied Statistics and Probability for Engineers**, 5th Edition, John Wiley and Sons, 2009.
3. Robert H. Shumway and David S. Stoffer, **Time Series Analysis and its Applications with R Examples**, 3rd edition, Springer Texts in Statistics, 2006.